## Research activity

**Controlling a calculation error — 30 min.**

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Denote by $e_n$ the error after $n$ steps, for $n \in \mathbb{N}$.

1. Express $e_n$ in terms of $n$.
2. Using the logarithm, determine the smallest integer $n$ such that $e_n \leq 10^{-4}$.
3. Using a calculator, calculate the error at the value found and at the previous value to check that it is minimal.
4. We now require an error of at most $10^{-6}$. Determine the new minimum number of steps. Does dividing the tolerance by 100 multiply the number of steps by 100?